Programming(3)

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Learning Objectives

- ✓ Understand what a loop is and why it’s needed
- ✓ Use for to repeat tasks
- ✓ Generate sequences with range()
- ✓ Check if an item exists with in / not in
- ✓ Write clear decision logic using match-case



match-case

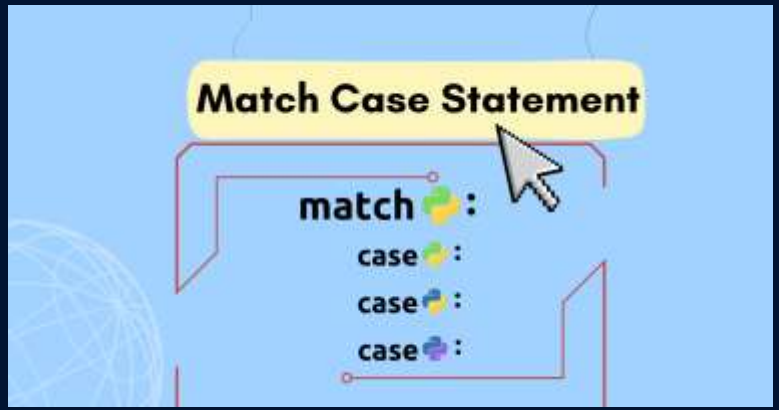


Match_Case

Introduced in Python 3.10+

Cleaner alternative to many if-elif statements.

- ✅ Always include case _: as the “default” option
- ✅ Case values must be exact (match is case-sensitive)
- ✅ Works great when checking fixed categories (e.g., days, menu options)



Match_Case

```python
color = input("Enter a color: ")

match color:
    case "red":
        print("Stop!")
    case "green":
        print("Go!")
    case "yellow":
        print("Wait!")
    case _:
        print("Invalid color.")

```



Membership Operators



Membership Operators

Used to check if something exists in a list, tuple, or string

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Membership Operators

Operator	Description	Example (Try in Lab)

in	Returns `True` if the variable/value is found in the specified sequence and `False` otherwise	`>>> a = [1,2,3]`
		`>>> 2 in a`
		`True`
		`>>> '1' in a`
		`False`

not in	Returns `True` if the variable/value is not found in the specified sequence and `False` otherwise	`>>> a = [1,2,3]`
		`>>> 10 not in a`
		`True`
		`>>> 1 not in a`
		`False`

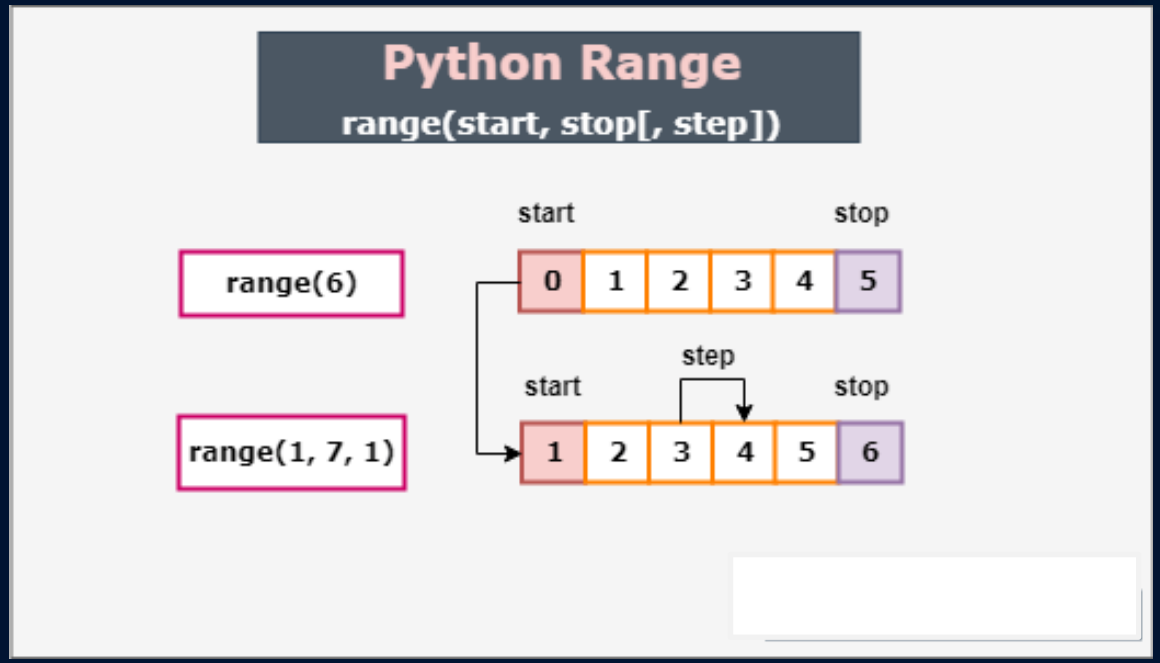


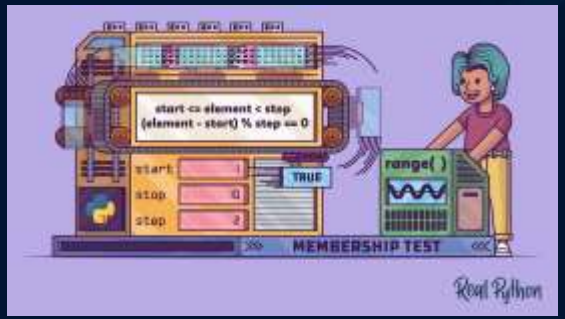
Range



Range

Creates a sequence of numbers





For

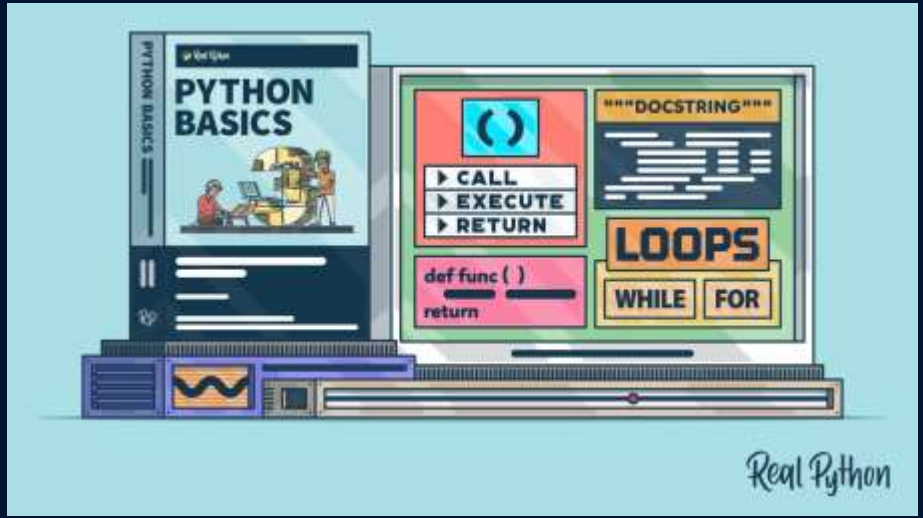


For

What is a Loop?

A loop lets us repeat a block of code several times.

Instead of writing the same line many times, we can use a loop.



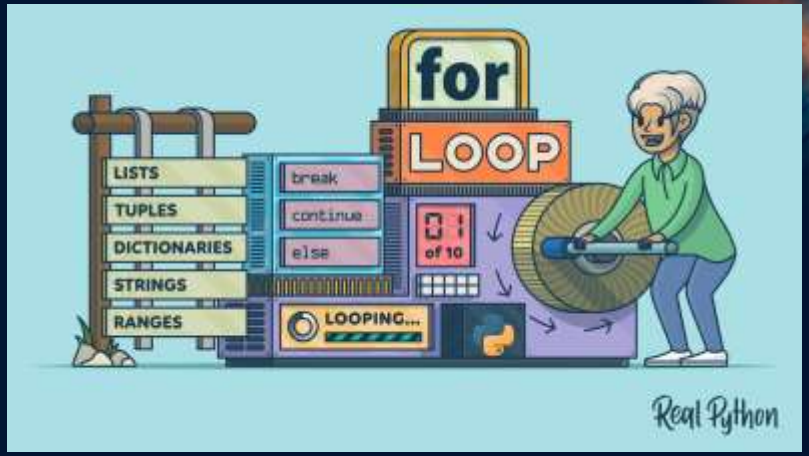
For

index.py > ...

```python
#   for variable in sequence:
#       code to run each time

for i in range(5):
    print("Number:", i)

'''
Number: 0
Number: 1
Number: 2
Number: 3
Number: 4
'''
```



Thank You

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